National University of Computer and Emerging Sciences
Peshawar Campus

Operating Systems (CS 2006)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 5

Date: 10th April 2026

Course Instructor(s)

Dr. Usman Ali Shah

Roll No ______ Section ______ Student Signature ______

Any use of calculators is STRICTLY PROHIBITED.

Do not write below this line.

Attempt all the questions.

CLO 1: Understand the characteristics of different structures of the Operating Systems and identify the core functions of the Operating Systems.

Q1: Give short answers (***seven or less lines***) to the following questions: **[Marks: 3*3=9]**

a. What are the different possibilities when we ***fork*** a new process from one of the threads of a multithreaded process?

b. Why does Pietersen's software-based synchronization solution fail on some modern processors? What can we do to avoid this failure?

c. Briefly explain the parameters, returned value and operations performed by the function ***test_and_set()***.

CLO 2: Analyze and evaluate the algorithms of the core functions of the Operating Systems and explain the major performance issues with regard to the core functions.

Q2: Consider the arrival times, CPU burst times and priorities for the following four processes. **[Marks: 8]**

Process	Arrival Time	CPU Burst Time	Priority
P_1	0	5	3
P_2	2	4	1
P_3	2	5	1
P_4	3	5	2

Build a Gantt Chart to depict the execution order of these four processes for a Preemptive Priority scheduling algorithm with Round-Robin scheduling (q = 2) for processes with the same priority. Also calculate the average waiting time and average response time of these processes.

Q3: Suppose that we have a program with a total of 50 instructions. Out of these 50 instructions, only 20 instructions are parallelizable. Also suppose that each instruction takes an equal amount of time to execute. Using this information, calculate the upper limit on speedup (for an infinite number of processors) against a single processor under Amdahl's Law to run this program. **[Marks: 3]**

CLO 3: Demonstrate the knowledge in applying system software and tools available in modern operating systems.

Q4: Consider the following code which attempts to provide complete synchronization among an *n* number of concurrent processes. **[Marks: 8]**

```
1.  // Shared Variables
2.  boolean waiting[n];          // All false, initially
3.  int lock = 0;
4.
5.  // Code for the i-th Process
6.  while(true)
7.  {
8.      waiting[i] = true;       // Raise hand: I "want" to enter my CS
9.      key = 1;
10.     while(waiting[i] == true && key == 1)
11.             key = compare_and_swap(&lock, 1, 0);
12.     waiting[i] = false;
13.
14.     // **** Critical Section ****
15.
16.     j = i++;
17.     while(j == i && waiting[j] == false)
18.             j++;
19.
20.     if (i == j)
21.             lock = 1;
22.     else
23.     waiting[j] = false;
24.
25.     // **** Remainder Section ****
26.     }
```

You are required to identify the errors in the above code that violate the synchronization requirements. Important: Just write down the number of the line with the error followed by its corrected version and the reason for the failure.

Q5: How is thread local storage different from ordinary local variables. **[Marks: 2]**

~~~~ Good Luck ~~~~
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National University of Computer and Emerging Sciences

Peshawar Campus

Database Systems (CS2005)

Date: April 13th, 2026

Course Instructors

Shoaib Mohammad Khan, Sanaa Jeehan

Sessional-II Exam

Total Time (Hrs):	**1**
Total Marks:	**35**
Total Questions:	**3**

Roll No ______ Section BSE-4B ______ Student Signature ______

Do not write below this line

Attempt all the questions.

CLO #2: Design a conceptual model using (E)ER diagrams for an enterprise

Q1: You have been hired to design a database for **GreenWheel**, a city-wide bicycle rental service. Read the business requirements below and translate them into a professional Entity-Relationship Diagram.

Requirements:

1. **Stations:** Each station has a unique **Station ID, a Name,** and a **Total Capacity** (number of docks).
2. **Bicycles:** Every bicycle has a unique **Serial Number** and a **Model Type** (e.g., Electric, Standard). Each bicycle is currently assigned to exactly one station, but a station can house many bicycles.
3. **Users:** The system tracks users by their **Email Address (unique), Full Name,** and **Phone Number.**
4. **Rentals:** A user can rent a bicycle. We must record the **Start Time, End Time,** and **Total Cost** of each rental.
   - A rental involves exactly one user and exactly one bicycle.
   - A user can have many rentals over time, and a bicycle can be used for many rentals.
5. **Maintenance:** Mechanics perform repairs. A mechanic has a **Staff ID** and a **Specialty.**
   - A mechanic can repair multiple bicycles, and a bicycle might be repaired by multiple mechanics over its lifespan.
   - For every repair, we must track the **Date of Repair** and a **Status Note.**

[10 marks]

CLO #3: Develop a normalized relational design to remove anomalies in a set of relations

Q2: The following unnormalized table is used by a medical clinic to track equipment maintenance and supplier contacts.

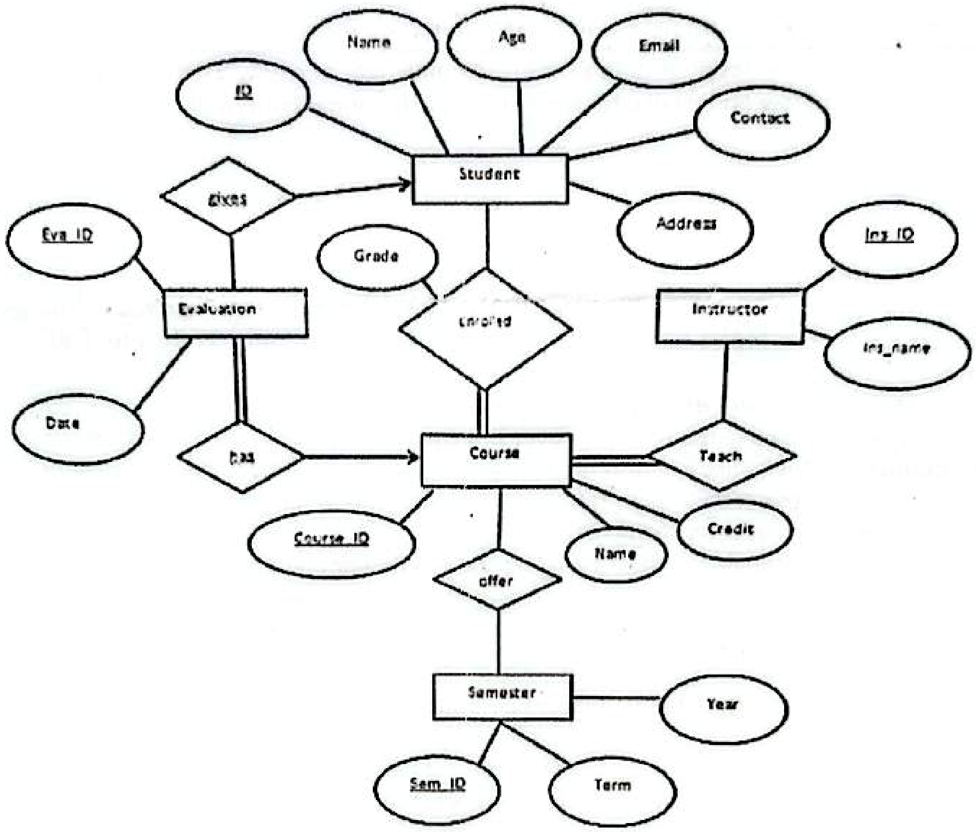
Item_ID	Item_Name	Supplier_ID	Supplier_Name	Supplier_City	Maintenance_Date	Maintenance_Cost
E101	MRI Scanner	S20	MedTech Inc.	Chicago	2024-01-12	500
E101	MRI Scanner	S20	MedTech Inc.	Chicago	2024-06-15	450
E205	X-Ray Mach.	S20	MedTech Inc.	Chicago	2024-02-10	200
E300	Ultrasound	S44	EchoLife	Austin	2024-03-05	150

Perform the following normalization steps. For each step, list the resulting tables, their attributes, and their Primary Keys (PK) and Foreign Keys (FK).

1. **First Normal Form (1NF):** Identify why this table might violate 1NF (if any multi-valued attributes existed) and ensure the table has a proper Primary Key.
2. **Second Normal Form (2NF):** Identify any partial functional dependencies and decompose the table to reach 2NF.
3. **Third Normal Form (3NF):** Identify any transitive dependencies and decompose the tables further to reach 3NF.

[15 marks]

Q3: Convert the following ERD into Tables with Primary Keys and Foreign Keys.
Note: arrow with relationship means 1, straight line means many and double line means total participation.



[10 marks]

National University of Computer and Emerging Sciences
Peshawar Campus

Software Requirements Engineering (SE2001)

Sessional II - Exam

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 2

Date: April 11 2026
Course Instructor(s)
Mr. Khizar Mukhtiar

Roll No   BSE-4B Section   Student Signature

Do not write below this line

Attempt all the questions.

CLO # 2: Effective Elicitation and Documentation

Consider the scenario below:

A construction company runs several projects at the same time. Currently each team keeps project information, such as schedules, purchase requests, site updates and payments, in different places. Management wants a single internal system so that project information is consistent and does not have to be entered in multiple places. The system is expected to support regular project work such as tracking planned work, recording site progress and issues, requesting materials, and monitoring project costs.

Each project also has a BIM model (a digital 3D building model) created by external consultants. The company receives updated model files when designs change. Teams want the new system to make the model useful during construction, for example by connecting planned work to parts of the building, attaching site observations to specific areas, and using model quantities to support purchasing. External consultants will continue sending model updates the way they do, they are not supposed to use the company's internal system. Material requests go out to suppliers and when finance approves invoices, payment is processed through the company's official bank account. In addition, finance also wants an accounting tool that records all financial reporting.

The business analyst is asked to describe the solution at a level that helps stakeholders agree on what the system will include, what it will connect to, and what the first release should focus on.

Q1. Answer the following questions:

a) For the scenario above, explain which scope representation techniques the BA would use to communicate the scope of Release 1 and the system boundary to stakeholders. Justify why each technique is appropriate in this scenario. **[5 marks]**
b) Choose any two of the techniques you mentioned in part (a) and draw them for this scenario. For each drawing, briefly explain the main design choices you made. **[10 marks]**

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CLO # 2: Effective Elicitation and Documentation

Consider the scenario below:

Pakistan Railways is launching a new online ticketing and journey management platform to replace several outdated systems. The platform must support millions of passengers across the country and will be used through a website, a mobile app, and station kiosks.

It must handle common passenger tasks such as searching routes, buying tickets, changing bookings, requesting refunds, and receiving disruption alerts. It also has to work with external systems such as payment providers, seat inventory, and the rail timetable system. Stakeholders include passengers, station staff, call-center agents, train conductors, finance, and security/compliance.

Q2. Answer the following questions:

a) As the business analyst, explain which requirements elicitation techniques you would use for this project and justify why they fit this scenario. Also explain who you would gather information from (name the main user groups and stakeholder representatives you would involve) and how you would practically collect the information from them. [7 marks]
b) After elicitation produces abundant information, the BA analyses it to derive user requirements. Explain the approaches the BA can take to arrive at user requirements, and the techniques used to express them. Elaborate on these techniques by describing how they would be applied in this project, including brief examples from the scenario, and state what the resulting outputs would be from each technique. [8 marks]

National University of Computer and Emerging Sciences

Peshawar Campus

Pakistan Studies (SS1003) — Sessional-II Exam

Date: April 09, 2026

Course Instructor(s)

Mr. Mubasher Basharat

Mr. Gohar Ali shah

Mr. Nadir Shah

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

Roll No ______ Section BSE-4B ______ Student Signature ______

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Attempt all the questions.

CLO #:3 *Analyze the socio-cultural diversity and inclusivity of Pakistan and understand how it helps in building a peaceful and united society.*

Q1: Explain the major impediments in the growth of democracy in Pakistan and discuss their impact on political development. **[10 marks]**

CLO #:4 *Evaluate Pakistan's economic development, foreign policy, and contemporary challenges, and assess their impact on national progress, governance, and sustainable development.*

Q2: Discuss the major problems that hinder economic development in Pakistan. **[10 marks]**

CLO #:4 *Evaluate Pakistan's economic development, foreign policy, and contemporary challenges, and assess their impact on national progress, governance, and sustainable development.*

Q3: What are the key factors that influence Pakistan foreign policy? **[10 marks]**

National University of Computer and Emerging Sciences
Peshawar Campus

OPERATING SYSTEM - LAB (CL - 2006)

Date: March 11th 2026

Lab Instructor(s):

Muhammad Saad Rashad

Lab MID Exam

Total Time (Hrs):	**1.15**
Total Marks:	**25**
Total Questions:	**2**

BSE-4B

Roll No.   Section   Student Signature

Instructions

1. Allowed tools: text editor (offline). Internet, communication devices and smart devices are strictly prohibited.
2. Submit all files created with the following format (***ROLL_QNO.format***).
3. Create a PDF/Word containing Screen Shots of output achieved.
4. ***Plagiarism*** will result in zero marks.

Attempt all the questions.

Q1) *File Name: RollNo._Q1.sh*

A small mobile recharge shop wants to calculate the total amount collected from customers in a day. Write a shell script that first asks the shopkeeper to enter the number of customers who purchased recharge cards. The script should then repeatedly ask the shopkeeper to enter the recharge amount for each customer. **[10 marks]**

Your script should perform the following tasks:

1. Display the recharge amounts that are greater than Rs. 500 (these are considered premium recharges).
2. Calculate and display the total amount collected from all customers.

Q2) *File Name: RollNo._Q2.c* **[15 Marks]**

A software company is developing a small Linux-based communication demo in C to illustrate inter-process communication using unnamed pipes. The program begins with a parent process that creates a child process. Before creating the child, the parent establishes an unnamed pipe so that both processes can exchange information. After the fork, the child process generates a short status message indicating that a background task has been completed and writes this message into the pipe. The parent process reads the message from the pipe and displays it on the screen along with the process IDs of both the parent and child processes to clearly demonstrate the communication relationship. Once the message has been successfully transmitted and displayed, the parent process waits for the child to terminate and finally prints "Communication Completed Successfully." Proper synchronization must be

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ensured so that the parent reads the data only after the child has written to the pipe and no orphan or zombie processes remain.